Apply for a Science Corps Fellowship - Entry #2389

Name

Chukwuma Nweke

Email

nwekec@umich.edu

Gender

Male

City, state, and country of residence

Ann Arbor

Nationality

United States

When did or will you finish your PhD?

August 2022

University Education

Doctor of Philosophy (PhD) in Macromolecular Science and Engineering, GPA 3.8 Aug. 2022
University of Michigan (UM), Ann Arbor, Michigan
Research Advisor: Dr. Jan P. Stegemann
Thesis title: Fabrication, Modification, and Evaluation of Modular, Gelatin-Based Microcarriers for the Delivery of Progenitor Cells in Bone Tissue Engineering Applications

Master of Science (MS) in Materials Science and Engineering, GPA 3.8 May 2015
Tuskegee University (TU), Tuskegee, Alabama

Bachelor of Science in Materials Science and Engineering, GPA 3.3 Dec. 2008
Georgia Institute of Technology (GT), Atlanta, Georgia

What are your reasons for applying to be a Science Corps fellow?

Please accept this letter as indication of my interest in seeking an available fellowship position within the Science Corps. I was recently awarded with a PhD in Macromolecular Science and Engineering, and am currently looking forward to my December 2022 graduation ceremony while looking to procure biomaterial research experience within dynamic research facilities.

My recent research experience as a graduate student in Dr. Jan Stegemann's Cell-Matrix Interactions and Tissue Engineering (CMITE) Lab helped me to conclude that developmental bench work is something I value very highly in a professional setting. This was indispensable to my stem cell and biopolymer-composite doctoral research. Throughout my term with the CMITE Lab, I was fortunate to explore firsthand human mesenchymal stem cell culture, composite synthetic/biopolymer syntheses, and various bioassay techniques. In an effort to explore potential injectability for resultant micro-tissue platforms, my work also incorporated thixotropic microbial transglutaminase- or nanoceramic-modified carrier gel formulation experiments, along with associated viscoelastic and thermomechanical testing. This followed master's level work in thermomechanical characterization of carbon nano-reinforced polyimide composites for the Air Force Research Lab (AFRL). Additionally, as an undergraduate researcher, I found the opportunity to work in photovoltaics fabrication, testing, and installation, in R&D for production window components, and on autonomic healing polymer composite research. In brief, while recognizing the compelling nature of your work, I hope that my technical experiences and ingenuity can act as an asset to other Science Corps fellows.

I'm eager to learn more about the available research projects from productive scientific leaders at Science Corps. And I hope that I'll be afforded the opportunity to join your dynamic team.

Thanks for taking the time to consider my application. I'm easily reachable by phone (770-286-0894) or email (nwekec@umich.edu).

Sincerely,

Chukwuma Nweke

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

UM: Cell-Matrix Interactions and Tissue Engineering (CMITE) Lab PhD Research 2017-2022
• Cultured, differentiated, imaged, and biologically assayed mesenchymal stem cells in modular microenvironment and delivery systems to explore the potential of a minimally invasive bone tissue engineering therapy
• Generated high-volume batches of gelatin-based microcarriers and hydrogels for morphological, rheological, degradation, and cytospecific characterization
• Synthesized and characterized photocrosslinkable poly(ethylene glycol) (PEG) diacrylamide, PEG-crosslinked collagen (PEG-COL), collagen, gelatin, nanosilicate, and hydroxyapatite-based biomaterials as tissue engineering scaffolds

TU/Air Force Research Laboratory: High-Temperature Composite MS Research 2014-2015
• Evaluated the thermomechanical effects of carbon nanotube (CNT) and other matrix modifiers on high-temperature thermoset polymer for F-22 airframe elements
• Increased glass transition temperature by 8.7%, storage modulus 23.28%, flexural strength 15.7%, flexural modulus 29.4%, and thermal stability by 20-24% in 0.15 wt.% CNT carbon fiber-reinforced thermosetting polyimide composites

GT: University Center of Excellence in Photovoltaics (UCEP) Undergraduate Research Jun.-Dec. 2008
• Produced and tested single p-n junction diode solar cell modules with energy conversion efficiencies of 17.5% for General Electric, Shell, etc.

• Administered H2SO4/HF etching, phosphorus (POCl3) diffusion, aluminum contact array printing, reflectance/resistivity measurements, and IV curve fill factor/efficiency measurements to polycrystalline silicon wafers/cells

GT: Surface Engineering and Molecular Assemblies (SEMA) Lab Senior Design Project 2007-2008
• Developed cupula-mimetic PEG and methyl methacrylate-based hydrogel deposition scheme for encapsulation of flow sensors in novel passive submarine detection method
• Preparatory oxidative and silanization treatment of silicon substrates for tailored coating of contoured hydrogel cupulae

GT: Solar Jackets Club Solar/Plugin Hybrid Vehicle Conversion Project Jun. Dec. 2008
• Managed solar cell roof array design division and solar ultraviolet light-based water purification system auxiliary project

GT: Student and Teacher Enhancement Partnership Aug. 2007 - May 2008
• Mentored, taught, and tutored students of Atlanta-based Tech High School in Math and Science
• Facilitated Georgia Tech RoboJackets & Tech High Robotics Club partnering program

Andersen Windows & Doors, Bayport, MN May - July 2007
• Modeled and ordered tooling for component prototypes in new window series launch

University of Illinois at Urbana-Champaign: Self-healing Polymer Internship Project May July 2006
• Fabricated, manually casted, and/or robotically extruded epoxy polymer blends embedded with urea-formaldehyde-encapsulated dicyclopentadiene healing agent and Grubb’s catalyst for extended life structural engineering
• Tested unbroken and crack-healed mechanical samples for 90% recovered fracture toughness

Where do you imagine yourself professionally in the future? Check all that apply.

Doing science in industry

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

In my immediate professional future, I hope to see myself in industrial R&D work. The biotechnology industry, which has some overlap with my past research experiences, is particularly varied. In the long-term, I'll be working to more broadly explore neighboring sub-fields of biotech.

CV/Resume

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How did you find out about Science Corps?

University email list

Any other comments/questions?

Empty

Science Corps